

Coverage Before Gluing: An Empty-Domain Audit for the Source-Backed Local Occurrence Family

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Abstract

TPC-175 extracts the largest production local occurrence-edge family supported by explicit theorem claims in the frozen TPC-133–172 corpus. That family is empty. We apply the TPC-165 local-to-global interface only to proved local edges and record two exact ledgers. Of 2,988 source-declared production cut addresses, zero are covered, zero duplicated, and 2,988 unmatched. Separately, there are zero eligible actual carriers and hence zero covered, duplicated, and unmatched eligible carriers. The unmatched cuts are not promoted to actual physical carriers, and neither ledger proves physical totality.

The nonempty-local-family hypothesis of the TPC-165 production gate is therefore unsatisfied, so its formal gluing conclusion is not invoked. The source-extraction cell is stopped in the exact frozen corpus scope, while the occurrence-augmented H1 architecture remains NOT-TESTABLE. The result is an L1 scoped interface obstruction, not an actual-support theorem, canonicity theorem, fixed-phase estimate, program-positive L2, or twin-prime result.

1 The admissible input

Let $\mathcal{E}_{175}$ be the maximal family certified by TPC-175. Membership requires a production claim to pass the source, locator, semantic, derivation, and witness-schema gates of TPC-173 and TPC-174. TPC-175 proves the scoped identity

$$|\mathcal{E}_{175}| = 0 \tag{1}$$

for the explicitly frozen corpus. Equation (1) is not a mathematical nonexistence theorem. It says that the declared source inventory contains no qualifying claim.

We define the eligible carrier domain by the carrier identifiers of proved edges:

$$\mathcal{C}_{\text{elig}} = \{c(e) : e \in \mathcal{E}_{175}\}. \tag{2}$$

Thus $\mathcal{C}_{\text{elig}} = \emptyset$. Definition (2) is deliberately narrow. It does not identify $\mathcal{C}_{\text{elig}}$ with the 2,988 frozen cut paths, nor with an unenumerated universe of physical carriers.

2 The TPC-165 gate

TPC-165 proves a finite descent theorem under supplied nonempty local row families and exact compatible overlap bijections [1]. The theorem is formal and conditional on those inputs. It does not construct them.

Proposition 2.1 (No production invocation on empty input). *The TPC-165 production gluing gate is not triggered by $\mathcal{E}_{175}$.*

Proof. The local nonemptiness hypothesis fails by (1). Applying the set-theoretic quotient operation to an empty family would yield an empty set, but that tautology is not the conclusion of the TPC-165 production theorem and proves no coverage statement. We therefore record *not invoked*, rather than “glued totality with zero rows.” $\square$

This distinction prevents a familiar vacuity error:

$$\forall e \in \emptyset, P(e) \not\equiv \exists e, P(e). \quad (3)$$

3 Exact coverage ledgers

Let $\mathcal{K}_{\text{cut}}$ be the 2,988 source-declared production cut addresses imported by TPC-175. For $k \in \mathcal{K}_{\text{cut}}$, let $n(k)$ be the number of proved local-edge rows above k . The exact classes are

$$\begin{aligned} \mathcal{K}_{\text{cov}} &= \{k : n(k) = 1\}, \\ \mathcal{K}_{\text{dup}} &= \{k : n(k) > 1\}, \\ \mathcal{K}_{\text{miss}} &= \{k : n(k) = 0\}. \end{aligned} \quad (4)$$

TPC-175 computes

$$(|\mathcal{K}_{\text{cov}}|, |\mathcal{K}_{\text{dup}}|, |\mathcal{K}_{\text{miss}}|) = (0, 0, 2988). \quad (5)$$

These are cut addresses, not actual physical carriers.

For the separately declared eligible carrier domain (2), let $\mathcal{C}_{\text{cov}}, \mathcal{C}_{\text{dup}}, \mathcal{C}_{\text{miss}}$ be the analogous multiplicity classes. All three are empty.

Theorem 3.1 (TPC-176 coverage ledger). *The exact frozen-cut and eligible-carrier ledgers are*

$$\begin{aligned} (|\mathcal{K}_{\text{cov}}|, |\mathcal{K}_{\text{dup}}|, |\mathcal{K}_{\text{miss}}|) &= (0, 0, 2988), \\ (|\mathcal{C}_{\text{cov}}|, |\mathcal{C}_{\text{dup}}|, |\mathcal{C}_{\text{miss}}|) &= (0, 0, 0). \end{aligned} \quad (6)$$

No production totality conclusion follows.

Proof. Equation (1) gives $n(k) = 0$ for all 2,988 declared cuts, proving the first line of (6). It also makes (2) empty, proving the second line. The final sentence follows because a cut address has not been proved to be an actual occurrence carrier, and the physical carrier universe has not been asserted to equal either declared domain. $\square$

Quantity	Count	Interpretation
declared production cuts	2988	source-address universe of TPC-175
covered cuts	0	exactly one proved local edge above the cut
duplicated cuts	0	more than one proved local edge above the cut
unmatched cuts	2988	no proved local edge; not an actual-carrier claim
eligible actual carriers	0	carrier identifiers in proved edges

4 Scoped stop versus architecture status

The extraction-and-gluing cell has exhausted its declared input:

$$\text{STOP_SCOPED_EMPTY_PROVED_LOCAL_EDGE_FAMILY}. \quad (7)$$

Its scope is the explicit TPC-133–172 source corpus and its declared data substrate. Enlarging that corpus, or proving a new source claim, creates a new audit instance.

The selected occurrence-augmented architecture is not stopped by (7). It may still be supplied by new mathematics or an explicitly enlarged source inventory. Its current status remains

$$\text{NOT_TESTABLE}. \quad (8)$$

This separation is essential: a stopped extraction method cell is not a proof that the complete architecture is infeasible.

5 Reproducible audit and claim boundary

The standard-library audit source-locks the TPC-165 formal theorem and the TPC-175 maximal-family export, recomputes (2)–(6), and mutation-tests the boundary. It rejects a fabricated positive edge, invocation of the gluing theorem on empty input, promotion of the 2,988 unmatched cuts to actual physical carriers, a production-totality claim, and a global architecture stop. Hashes carry integrity semantics only.

TPC-176 consumes no literal physical coefficient, named phase, deterministic endpoint, normalization, or fixed- $h_0 = 2$ arithmetic estimate. Consequently it supplies no actual active support, canonical physical representative, program-positive L2, or strict 1/400 endpoint gain.

6 Conclusion

The correct output of an empty proved local-edge family is an exact empty-domain ledger plus a scoped stop, not a vacuous totality certificate. TPC-176 closes the frozen-corpus extraction cell while leaving all three H1 structural roots visible. TPC-177 next asks what an actual active-support audit can and cannot say when its eligible carrier domain is empty.

References

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